

PROGRAM BOOK FOR COMMUNITY SERVICE PROJECT

Name of the Student: Narendra Kumar Paswan

Name of the College: Sri Venkateswara College of Engineering and Technology
(Autonomous), Chittoor

Department: Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Registration Number: 24781A33J7

Period of CSP: From: 10 May 2026 To: 29 June 2026

Name & Address of the Community / Habitation: Peddapanjani Village, Chittoor
District, Andhra Pradesh

Group No.: 16

Faculty Guide: Mrs. T. Amruthavalli

Jawaharlal Nehru Technological University, Ananthapuramu

Academic Year 2024–2028

COMMUNITY SERVICE PROJECT REPORT

Submitted in accordance with the requirement for the degree of
Bachelor of Technology (B.Tech)

Name of the College: Sri Venkateswara College of Engineering and Technology
(Autonomous), Chittoor

Department: Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Name of the Faculty Guide: Mrs. T. Amruthavalli

Duration of the CSP: From 7 May 2026 to 29 June 2026

Name of the Student (Group Leader): Narendra Kumar Paswan — Reg. No. 24781A33J7

Name of the Student (Member 2): Govinda Aryal — Reg. No. 24781A33J3

Name of the Student (Member 3): Sumit Aryal — Reg. No. 24781A33F9

Name of the Student (Member 4): Mekala Anil Kumar — Reg. No. 24781A33A6

Name of the Student (Member 5): Abhishek Shah — Reg. No. 23781A33H7

Programme of Study: B.Tech – CSE (Artificial Intelligence & Machine Learning)

Year of Study: III Year B.Tech

Date of Submission: 30 June 2026

STUDENT'S DECLARATION

I, Narendra Kumar Paswan, a student of B.Tech – CSE (AI & ML) Programme, Reg. No. 24781A33J7 of the Department of Computer Science and Engineering (Artificial Intelligence & Machine Learning), Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor do hereby declare that I have completed the mandatory community service from 10 May 2026 to 29 June 2026 in Peddapanjani Village, Chittoor District, Andhra Pradesh under the Faculty Guideship of Mrs. T. Amruthavalli, Department of Computer Science and Engineering (AI & ML) in the College.

I further declare that the work presented in this report is original and carried out by our team during the stipulated period of Community Service. No part of this report has been submitted elsewhere for the award of any degree or diploma.

Narendra Kumar Paswan (Group Leader)

Endorsements:

_____ Mrs. T. Amruthavalli (Faculty Guide)

_____ Head of the Department, CSE (AI & ML)

_____ Principal, SVCET (Autonomous), Chittoor

CERTIFICATE FROM OFFICIAL OF THE COMMUNITY

This is to certify that Narendra Kumar Paswan (Group Leader) and team (Group No. 16), Reg. No. 24781A33J7 and team of Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor underwent community service in Peddapanjani Village, Chittoor District, Andhra Pradesh from 10 May 2026 to 29 June 2026.

The team conducted a socio-economic survey of the community, organized a healthcare awareness programme, and actively engaged with residents and local officials throughout the duration of the project. The overall performance of the Community Service Volunteers during their community service is found to be

GOOD

The team demonstrated professionalism, dedication, and a genuine commitment to community welfare throughout the project period.

Name of Authorized Signatory: Mr. K. Ravi Kumar

Designation: Village Administrative Officer (VAO), Peddapanjani Village

Date: 21 June 2026

Mr. K. Ravi Kumar Authorized Signatory with Seal

ACKNOWLEDGEMENT

We express our sincere gratitude to our Faculty Guide, Mrs. T. Amruthavalli, for her unwavering support, expert guidance, and continuous encouragement throughout the duration of this Community Service Project. Her timely feedback and mentorship were invaluable in shaping the direction and quality of our work.

We are deeply grateful to the Principal and the Head of the Department of Computer Science and Engineering (AI & ML) at Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor for providing us with the opportunity to undertake this meaningful project as part of our academic programme.

We extend our heartfelt thanks to Mr. K. Ravi Kumar, Village Administrative Officer of Peddapanjani Village, Chittoor District, Andhra Pradesh, for his cooperation and for facilitating our engagement with community members, leaders, and officials. We thank the Panchayat President, the Auxiliary Nurse Midwife (ANM) at the local health sub-centre, and the school headmaster for generously sharing their time and insights.

We are grateful to the residents of Peddapanjani Village, Chittoor District, Andhra Pradesh for their warmth, patience, and willingness to participate in our survey and awareness activities. Their cooperation made this project both productive and deeply rewarding.

We also thank the Andhra Pradesh State Council of Higher Education (APSCHE) for designing a structured Community Service Project framework that connects academic learning with real-world community engagement.

Team Members:

- Narendra Kumar Paswan — Reg. No. 24781A33J7
- Govinda Aryal — Reg. No. 24781A33J3
- Sumit Aryal — Reg. No. 24781A33F9
- Mekala Anil Kumar — Reg. No. 24781A33A6
- Abhishek Shah — Reg. No. 23781A33H7

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CHAPTER 1: EXECUTIVE SUMMARY

This report presents the work carried out during the Community Service Project (CSP) undertaken from 10 May 2026 to 29 June 2026 in Peddapanjani Village, Chittoor District, Andhra Pradesh, as part of the mandatory 10-month internship component of the B.Tech – CSE (AI & ML) programme at Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor. The five-member CSP team (Group No. 16), led by Narendra Kumar Paswan, worked under the guidance of Mrs. T. Amruthavalli.

The team engaged with Peddapanjani village — a semi-rural community in Chittoor District — to understand its socio-economic conditions, identify pressing community problems, and propose technology-based solutions. The team conducted a structured household-level socio-economic survey covering 68 households, interacted with community leaders and the local health worker, and organized a healthcare awareness programme attended by approximately 65 residents on 09 June 2026.

A mini-project titled 'Mobile App for Rural Healthcare Assistance' was developed as the technical component of the CSP, directly informed by the healthcare access gaps and resident feedback gathered during field engagement. The application is designed to assist rural residents in managing personal health information, receiving medicine and appointment reminders, locating nearby healthcare facilities, and accessing basic health awareness content.

Learning Objectives and Outcomes:

1. Gained first-hand understanding of the socio-economic realities and healthcare challenges of a rural community through direct field engagement.
2. Developed professional communication and interpersonal skills through structured interaction with residents, community leaders, and officials.
3. Acquired practical skills in survey design, data collection, and community-level problem analysis.
4. Applied software engineering and mobile application design principles to a real-world, socially relevant problem, developing a conceptual healthcare app aligned with identified community needs.
5. Strengthened teamwork, project planning, technical documentation, and report-writing skills through a sustained six-week collaborative effort.

CHAPTER 2: OVERVIEW OF THE COMMUNITY

2.1 Historical Profile

Peddapanjani is a small semi-rural village located in Chittoor District, Andhra Pradesh, approximately 18 kilometres from the district headquarters. The village has a recorded history of agricultural settlement spanning several generations, with residents traditionally engaged in paddy, groundnut, and tomato cultivation. The village falls under the Chittoor Revenue Division and is governed locally by an elected Gram Panchayat. Key community institutions include a primary school, an anganwadi centre, and a primary health sub-centre staffed by an Auxiliary Nurse Midwife (ANM).

In recent decades, improved road connectivity and electrification have gradually changed the living patterns of the community, with a growing number of younger residents pursuing secondary and higher education in nearby towns. The village has a strong tradition of community cohesion, with festivals such as Sankranti, Ugadi, and local temple events serving as focal points of social bonding.

2.2 Community Diversity, Traditions, and Values

Peddapanjani is home to a predominantly Telugu-speaking population representing multiple social and occupational groups, including agricultural families, daily-wage labourers, and a small number of small traders. Community values emphasize mutual respect, collective participation in village decision-making through Panchayat meetings, and care for elders. Traditional practices such as community water management and cooperative farming remain active.

2.3 Socio-Economic Conditions

Based on the socio-economic survey conducted by the team across 68 households, the following conditions were documented:

- Primary occupation: approximately 60% of households depend on agriculture, 25% on daily labour, and 15% on small trade or services.
- Average household size: 4.2 members.
- Educational attainment: most adults aged above 40 have completed primary education; younger residents increasingly complete secondary schooling.
- Access to electricity: available in approximately 94% of households.
- Access to safe drinking water: piped water supply available; reliability varies by season.
- Access to healthcare: the nearest government hospital is approximately 12 kilometres away; the local health sub-centre provides limited services.

- Mobile phone ownership: approximately 78% of households own at least one mobile phone; smartphone penetration is approximately 52%.

The relatively high mobile phone penetration identified in the survey presents a significant opportunity for digital healthcare tools to reach and benefit the community.

CHAPTER 3: COMMUNITY SERVICE PART

During the six-week Community Service Project at Peddapanjani village, the team undertook a structured and progressive series of activities designed to understand, engage with, and support the community. Activities included initial community orientation, systematic household-level data collection through a structured socio-economic survey, analysis of survey findings, planning and conducting a community health awareness programme, and developing a technology-based mini-project directly informed by community needs.

The team maintained daily activity logs signed by the community person in-charge and submitted weekly reports to the Faculty Guide. Team members were present for all six working days each week, wore College identity cards throughout, and maintained professional conduct consistent with the guidelines prescribed by APSCHE and the College.

Reflection — Values, Life Skills, and Technical Skills Acquired:

- Empathy and sensitivity while engaging with households from diverse socio-economic backgrounds.
- Improved oral and written communication skills in both English and Telugu through sustained community interaction.
- Enhanced problem-identification and analytical thinking through structured survey data analysis.
- Practical experience in survey design, data collection, and community-level programme organization.
- Applied technical skills in software architecture, database design, and mobile application planning.
- Strengthened teamwork, coordination, and project management abilities across a six-week collaborative project.
- Developed a deeper appreciation for the social responsibilities of engineers and the role of technology in community development.

ACTIVITY LOG — WEEK 1

Period: 10 May 2026 – 15 May 2026

Day	Date	Brief Description of Activity	Learning Outcome	Signature
Day 1	10 May 2026	Orientation: reported to VAO, met community leaders, initial village walkthrough.	Understood village layout and key community institutions.	
Day 2	12 May 2026	Interacted with Panchayat President and discussed village development priorities.	Identified healthcare and infrastructure as the community's top concerns.	
Day 3	13 May 2026	Visited the primary health sub-centre; spoke with the ANM about patient attendance trends.	Understood the limited capacity of local health infrastructure.	
Day 4	14 May 2026	Visited the village school; interacted with teachers and students about health awareness levels.	Recognized low awareness of preventive healthcare among younger population.	
Day 5	15 May 2026	Informal household visits across three streets; noted occupations and household composition.	Gained first-hand understanding of livelihoods and daily life of residents.	
Day 6	16 May 2026	Team meeting to synthesize observations; began drafting socio-economic survey questionnaire.	Developed a structured approach to data collection for the following week.	

WEEKLY REPORT — WEEK 1

Period: 10 May 2026 – 15 May 2026

Objective of the Activity Done:

To introduce the team to the community, build initial rapport with residents and local leaders, and begin preliminary observation of the village's socio-economic environment.

Detailed Report:

The team arrived at Peddapanjani village on Day 1 and reported to the community person in-charge, Mr. K. Ravi Kumar (Village Administrative Officer). The Faculty Guide, Mrs. T. Amruthavalli, accompanied the team for the first day's orientation. The team was formally introduced to community leaders including the Panchayat President, the local health worker (ANM), and the headmaster of the village primary school. An initial walkthrough of the village helped the team understand its layout, identify key institutions such as the anganwadi centre, primary health sub-centre, and local wells. Informal conversations with residents across all six days helped build trust, understand primary occupations, and identify preliminary concerns regarding healthcare access. The team also established a WhatsApp group with the Faculty Guide for daily updates and maintained individual activity logs from Day 1.

ACTIVITY LOG — WEEK 2

Period: 19 May 2026 – 24 May 2026

Day	Date	Brief Description of Activity	Learning Outcome	Signature
Day 1	19 May 2026	Finalized questionnaire; pilot survey with 5 households; refined 2 ambiguous questions.	Improved clarity and reliability of the survey instrument.	
Day 2	20 May 2026	Full survey: visited 10 households in the eastern section of the village.	Collected data on occupation, income, and healthcare access patterns.	
Day 3	21 May 2026	Survey continued: 10 households in the western section; noted higher smartphone ownership than expected.	Identified potential for mobile-based digital healthcare solution.	
Day 4	22 May 2026	Survey of 8 households near the anganwadi; spoke with mothers about child health and nutrition.	Observed gaps in maternal and child healthcare awareness.	
Day 5	23 May 2026	Survey: 7 households of elderly residents; noted difficulty with medicine adherence.	Identified medicine reminder feature as a high-priority app requirement.	
Day 6	24 May 2026	Compiled survey responses; prepared preliminary data summary for review by Faculty Guide.	Organized raw data into structured format for analysis.	

WEEKLY REPORT — WEEK 2

Period: 19 May 2026 – 24 May 2026

Objective of the Activity Done:

To finalize the socio-economic survey questionnaire, pilot it with selected households, and begin full-scale household data collection.

Detailed Report:

The team finalized a structured questionnaire covering eight domains: household demographics, primary occupation, income sources, education, access to drinking water, access to electricity and sanitation, healthcare access, and mobile/smartphone usage. The questionnaire was piloted with five households on Day 1 of the week; minor adjustments were made to simplify two questions that respondents found ambiguous. From Day 2 onward, the team divided into two pairs and systematically covered different sections of the village. By Day 6, approximately 40 households had been surveyed. The team compiled raw responses each evening and posted daily summaries in the Faculty Guide's WhatsApp group. The survey was conducted with courtesy and care, ensuring no disruption to household activities.

ACTIVITY LOG — WEEK 3

Period: 26 May 2026 – 31 May 2026

Day	Date	Brief Description of Activity	Learning Outcome	Signature
Day 1	26 May 2026	Completed survey: remaining 8 households; total sample reached 68 households.	Completed primary data collection for socio-economic analysis.	
Day 2	27 May 2026	Data tabulation: organized survey responses into category-wise tables.	Structured data enabled identification of clear demographic and health patterns.	
Day 3	28 May 2026	Preliminary analysis: identified 5 key healthcare problems in the community.	Problem identification gave focus and direction to the awareness programme.	
Day 4	29 May 2026	Met ANM and headmaster to discuss awareness programme format and venue.	Confirmed venue (school hall) and scheduled programme for Week 5.	
Day 5	30 May 2026	Designed content for awareness programme: hygiene, preventive care, medicine adherence.	Prepared a structured and practical awareness session plan.	
Day 6	31 May 2026	Submitted written data analysis summary to Faculty Guide for review and feedback.	Received validation and suggestions to strengthen the awareness programme content.	

WEEKLY REPORT — WEEK 3

Period: 26 May 2026 – 31 May 2026

Objective of the Activity Done:

To complete household data collection, conduct preliminary analysis of survey findings, and identify key community problems.

Detailed Report:

The third week focused on completing the survey across remaining households and beginning data analysis. By Day 2, the survey covered 68 households in total, representing approximately 85% of the targeted sample. The team consolidated all responses and identified five primary problem areas: (1) limited proximity to healthcare facilities, (2) low awareness of preventive healthcare, (3) irregular medication adherence among elderly residents, (4) lack of a central source of basic health information, and (5) limited awareness of government healthcare schemes. Findings were presented informally to the Faculty Guide via a written summary. The team also began planning the community awareness programme for Week 5, consulting the ANM and local school headmaster about an appropriate date and venue.

ACTIVITY LOG — WEEK 4

Period: 02 June 2026 – 07 June 2026

Day	Date	Brief Description of Activity	Learning Outcome	Signature
Day 1	02 Jun 2026	Developed Telugu visual aids on hygiene and preventive health for the awareness programme.	Improved communication effectiveness by using the local language.	
Day 2	03 Jun 2026	Prepared content on government health schemes (Ayushman Bharat, JSY) for the awareness session.	Gained knowledge of key government healthcare entitlements for rural residents.	
Day 3	04 Jun 2026	Conducted team rehearsal for awareness programme; incorporated Faculty Guide's feedback.	Improved clarity, structure, and confidence of the presentation.	
Day 4	05 Jun 2026	Visited anganwadi: interacted with health worker and mothers on child nutrition and immunization.	Strengthened understanding of maternal and child health issues in the community.	
Day 5	06 Jun 2026	Distributed awareness about the upcoming programme to households; confirmed attendance.	Ensured broad community participation in the planned awareness session.	
Day 6	07 Jun 2026	Final preparation: printed handouts; arranged school hall with Faculty Guide's assistance.	Programme logistics finalized and team fully prepared for the event.	

WEEKLY REPORT — WEEK 4

Period: 02 June 2026 – 07 June 2026

Objective of the Activity Done:

To prepare and rehearse the community awareness programme and engage residents in preliminary health discussions.

Detailed Report:

The fourth week was dedicated to thorough preparation for the awareness programme. The team developed visual aids in the local language (Telugu) covering four topics: hand hygiene and sanitation, preventive healthcare practices, the importance of regular medical check-ups, and correct medication usage. A rehearsal session was conducted on Day 3 to refine the presentation and improve clarity. The team also continued community engagement by visiting the anganwadi centre and interacting with the health worker about ongoing government schemes such as Ayushman Bharat and the Janani Suraksha Yojana, noting that many residents were unaware of their entitlements. These interactions enriched the awareness programme content significantly.

ACTIVITY LOG — WEEK 5**Period: 09 June 2026 – 14 June 2026**

Day	Date	Brief Description of Activity	Learning Outcome	Signature
Day 1	09 Jun 2026	Conducted community health awareness programme (65 attendees); 4 sessions with Q&A.	Delivered preventive healthcare information to a wide cross-section of the community.	
Day 2	10 Jun 2026	Post-programme visit: collected verbal feedback from 12 households on session usefulness.	Confirmed positive reception; residents valued the information on government schemes.	
Day 3	11 Jun 2026	Follow-up with elderly residents; demonstrated basic smartphone health features.	Identified practical interest in mobile-based medicine reminders among elderly users.	
Day 4	12 Jun 2026	Discussed programme outcomes with the ANM; shared summary of participant feedback.	Established community-level record of awareness programme and its impact.	
Day 5	13 Jun 2026	Began drafting mini-project report based on community findings and programme outcomes.	Translated community observations into structured technical documentation.	
Day 6	14 Jun 2026	Team discussion: finalized mini-project features based on direct community feedback.	Aligned application design with actual user needs identified in the field.	

WEEKLY REPORT — WEEK 5

Period: 09 June 2026 – 14 June 2026

Objective of the Activity Done:

To conduct the community health awareness programme and document its outcomes and impact.

Detailed Report:

The community awareness programme was conducted on Day 1 (09 June 2026) at the village primary school hall, attended by approximately 65 residents including women, farmers, youth, and elderly community members, along with the VAO and the ANM. The programme comprised four sessions: (1) hand hygiene and sanitation practices, (2) preventive healthcare and regular check-ups, (3) correct use of prescribed medication and the importance of adherence, and (4) an overview of government healthcare schemes available to rural residents. An interactive Q&A segment generated enthusiastic participation. Multiple residents expressed interest in a mobile-based tool for medicine reminders and health information, directly validating the team's mini-project concept. Post-programme household follow-up visits on Days 2–6 allowed the team to collect feedback and reinforce key messages.

ACTIVITY LOG — WEEK 6

Period: 16 June 2026 – 21 June 2026

Day	Date	Brief Description of Activity	Learning Outcome	Signature
Day 1	16 Jun 2026	Finalized mini-project system architecture and database design documentation.	Produced a complete technical design for the rural healthcare mobile application.	
Day 2	17 Jun 2026	Completed mini-project report including technology stack and future scope sections.	Demonstrated ability to translate community needs into a full software design document.	
Day 3	18 Jun 2026	Closing visit to VAO and Panchayat President; shared summary of findings and recommendations.	Formally communicated CSP outcomes to community leadership.	
Day 4	19 Jun 2026	Thank-you visit to ANM and school headmaster; obtained community certificate.	Received formal acknowledgement of community service from the person in-charge.	
Day 5	20 Jun 2026	Compiled all activity logs, weekly reports, survey data into the final report draft.	Developed comprehensive documentation and report-writing skills.	
Day 6	21 Jun 2026	Submitted final report draft to Faculty Guide; incorporated final feedback and corrections.	Completed the Community Service Project with all required deliverables.	

WEEKLY REPORT — WEEK 6

Period: 16 June 2026 – 21 June 2026

Objective of the Activity Done:

To finalize the mini-project design, consolidate all CSP deliverables, and conduct closing interactions with the community.

Detailed Report:

The final week of the Community Service Project was dedicated to consolidating all deliverables and formally concluding engagement with the community. The team finalized the mini-project report, including system architecture, database design, technology stack, and future scope sections. Closing visits were made to the community person-in-charge, Panchayat President, ANM, and school headmaster to thank them for their cooperation and to share a brief summary of the team's findings and recommendations. The VAO acknowledged the team's efforts and expressed appreciation for the awareness programme. All activity logs, weekly reports, survey data, and mini-project content were compiled into the final CSP report and submitted to the Faculty Guide for review before formal submission.

CHAPTER 4: OUTCOMES DESCRIPTION

4.1 Socio-Economic Survey of the Village

A structured questionnaire comprising 28 questions across eight thematic sections was administered to 68 households in Peddapanjani village. The survey was conducted in Telugu (the community's primary language) with assistance from local residents where required. The key findings are summarized below:

Domain	Key Finding
Household Composition	Average household size: 4.2 members; 42% have 5+ members.
Primary Occupation	Agriculture: 60%; Daily Labour: 25%; Trade/Services: 15%.
Monthly Income	Approximately 55% of households earn below ₹10,000 per month.
Education	74% of adults aged 15–35 have completed at least 10th standard.
Electricity Access	94% of households have electricity supply.
Safe Drinking Water	Piped supply available; 18% report seasonal shortages.
Healthcare Access	Nearest hospital: ~12 km; 68% depend on the local ANM sub-centre.
Mobile/Smartphone	78% own a mobile phone; 52% own a smartphone.

4.2 Problems Identified

Based on the survey data and direct community interaction, five primary problems were identified:

6. Limited access to healthcare facilities — the nearest government hospital is approximately 12 kilometres away, making travel difficult for elderly residents and those without transport.
7. Low awareness of preventive healthcare — many residents lack knowledge of preventive practices such as hand hygiene, balanced diet, and the importance of regular health check-ups.
8. Irregular medication adherence — particularly among elderly residents managing chronic conditions, with no reliable reminder system in place.

9. Absence of a centralized health information source — residents rely on informal networks for health-related information, which is often inaccurate or incomplete.
10. Limited awareness of government healthcare schemes — many households are unaware of schemes such as Ayushman Bharat and Janani Suraksha Yojana to which they are entitled.

4.3 Short-Term and Long-Term Action Plan

Short-Term Recommendations (0–6 months):

- Conduct periodic health camps in the village in coordination with the District Health Officer.
- Distribute printed materials in Telugu on preventive healthcare, hygiene, and medicine adherence.
- Facilitate enrollment of eligible households in government healthcare schemes with assistance from the local ANM.

Long-Term Recommendations (6 months – 3 years):

- Establish regular staffing at the local health sub-centre to reduce dependence on distant hospitals.
- Deploy a mobile-based healthcare assistance application (as proposed in the mini-project) to provide digital health information, reminders, and facility locator services to residents.
- Partner with educational institutions for ongoing digital literacy initiatives to support the adoption of digital healthcare tools.
- Advocate for improved internet connectivity infrastructure to support digital health services.

4.4 Community Awareness Programme Report

Date of Programme: 09 June 2026

Venue: Village Primary School Hall, Peddapanjani

Number of Attendees: Approximately 65 residents

Key Attendees: Village residents (men, women, elderly, youth), VAO Mr. K. Ravi Kumar, ANM

Programme Sessions:

11. Hand hygiene and sanitation — demonstrated proper handwashing techniques and discussed the link between hygiene and disease prevention.
12. Preventive healthcare and regular check-ups — explained the importance of annual health check-ups, common preventable diseases, and early detection.
13. Correct medication use and adherence — addressed common myths about medicines, emphasized the importance of completing prescribed courses, and discussed safe storage.
14. Government healthcare schemes — explained eligibility and benefits of Ayushman Bharat, Janani Suraksha Yojana, and the National Health Mission for rural residents.

Outcomes:

- 65 residents attended, exceeding the team's target of 50 attendees.
- Post-session discussions indicated increased awareness of preventive healthcare and government scheme entitlements.
- Multiple residents, particularly elderly participants, expressed strong interest in a mobile tool for medicine reminders and health information.
- The ANM confirmed she would follow up with households on scheme enrollment in subsequent weeks.

MINI PROJECT REPORT

Mobile App for Rural Healthcare Assistance

Group No.: 16 | Faculty Guide: Mrs. T. Amruthavalli

5.1 Introduction

The 'Mobile App for Rural Healthcare Assistance' is a conceptual mobile application designed to address the healthcare accessibility and awareness challenges identified during the Community Service Project at Peddapanjani village. The application targets rural residents — particularly those with limited access to healthcare facilities — and is designed to assist them in managing personal health information, receiving timely reminders, locating nearby healthcare facilities, and accessing reliable health awareness content. The app concept was directly validated by community feedback during the CSP awareness programme, where multiple residents expressed need for exactly such a tool.

5.2 Problem Statement

Rural communities in Chittoor District face significant barriers to healthcare access: the nearest government hospital is approximately 12 kilometres away, local health infrastructure is limited, and awareness of preventive healthcare remains low. Despite smartphone penetration of approximately 52% in Peddapanjani, no dedicated digital healthcare tool exists that is simple enough for users with varying digital literacy levels and relevant to the specific needs of this rural community. The proposed application addresses this gap.

5.3 Objectives

15. Design a mobile application that allows users to create and maintain a basic digital health profile.
16. Provide configurable medicine and appointment reminders to improve treatment adherence.
17. Enable users to locate nearby hospitals, clinics, and pharmacies.
18. Provide curated basic health awareness content relevant to common rural health concerns.
19. Store emergency contact details for quick access in medical situations.
20. Design the application with a simple, accessible UI suitable for users with limited digital literacy.

5.4 Proposed Solution and Key Features

- **User Registration and Authentication:** Secure login with mobile number OTP or simple PIN for low-literacy users.
- **Digital Health Profile:** Name, age, blood group, allergies, chronic conditions, current medications.
- **Medicine Reminders:** Configurable daily/weekly reminders with custom medicine names and dosage notes.
- **Appointment Reminders:** Date and time-based reminders for upcoming medical consultations.
- **Emergency Contacts:** Quick-access list of emergency contacts and the nearest healthcare facility number.
- **Healthcare Facility Locator:** GPS-based map view of nearby hospitals, clinics, and pharmacies.
- **Health Awareness Content:** Curated short articles and infographics on hygiene, preventive care, and common diseases, available in Telugu and English.

5.5 System Architecture

The application follows a three-tier client-server architecture:

- **Presentation Tier:** Mobile front-end developed using Flutter (cross-platform, supports both Android and iOS), with a simple, high-contrast UI designed for varying screen sizes and digital literacy levels.
- **Application Tier:** A RESTful backend API built using Flask (Python), handling user authentication, health profile management, reminder scheduling, and content delivery.
- **Data Tier:** SQLite for local device storage (offline support); PostgreSQL for server-side persistent storage. Facility location data retrieved via Maps API.
- **External Integrations:** Google Maps API for facility locator; Firebase Cloud Messaging (FCM) for push notifications and reminders.

5.6 Use Case Overview

Primary Actors: Rural Resident (Patient/User), Healthcare Worker (optional secondary actor). Key use cases include: Register and Login, Create/Update Health Profile, Set and Manage Medicine Reminders, Set Appointment Reminder, View Nearby Healthcare Facilities, Access Health Awareness Content, Manage Emergency Contacts, and Receive Push Notification Reminders.

5.7 Database Design

Table	Primary Key	Key Fields	Purpose
Users	user_id	name, age, gender, phone, pin_hash	Stores registered user details.
HealthProfile	profile_id	user_id, blood_group, allergies, conditions, medications	Stores user health profile.
Reminders	reminder_id	user_id, type, label, time, frequency, active	Medicine/appointment reminders.
EmergencyContacts	contact_id	user_id, name, relation, phone	User emergency contacts.
HealthFacilities	facility_id	name, type, lat, lon, phone, distance	Nearby healthcare facility data.
AwarenessContent	content_id	title, category, body_en, body_te, created_date	Health awareness articles.

5.8 Technology Stack

Component	Technology / Tool	Justification
Mobile Front-End	Flutter (Dart)	Cross-platform; single codebase for Android and iOS; rich UI widgets.
Backend Framework	Flask (Python)	Lightweight, easy to extend; suitable for REST API development.
Local Database	SQLite	Offline-first support; no server needed for basic functionality.
Server Database	PostgreSQL	Reliable, scalable relational database for server-side storage.
Authentication	Mobile OTP / PIN	Simple, accessible for low-literacy and elderly users.
Notifications	Firebase Cloud Messaging	Reliable push notifications for medicine/appointment reminders.
Maps & Location	Google Maps API	Facility locator with GPS-based proximity search.
Version Control	Git / GitHub	Source code management and team collaboration.

5.9 Future Scope and AI Integration

The proposed Mobile App for Rural Healthcare Assistance is designed as a scalable, extensible solution. Future versions can incorporate advanced technologies — including Artificial Intelligence (AI), Machine Learning (ML), cloud computing, and IoT — to deliver smarter, more personalized healthcare services to rural communities.

1. AI-Based Symptom Checker

An AI-powered symptom checker can allow users to enter symptoms and receive guidance on possible conditions and whether to seek professional medical care. This feature must be clearly communicated as informational only — not a substitute for professional medical diagnosis.

2. Telemedicine Integration

Future versions can include video or audio consultation features connecting users with healthcare professionals, reducing the need for long-distance travel to access medical advice.

3. Cloud-Based Health Record Synchronization

A cloud storage layer can enable health records to be securely backed up and accessed across multiple devices, preventing data loss and improving continuity of care.

4. ML-Based Medicine Adherence Prediction

Machine learning models trained on anonymized adherence data can predict which users are at risk of missing medication doses and proactively send personalized reminders or alerts to caregivers.

5. Wearable Device Integration

Future versions can interface with affordable wearable devices to capture biometric data such as heart rate, step count, sleep patterns, and blood oxygen levels, feeding this data into the health profile for a more complete wellness picture.

6. Multilingual Support

Additional regional language support (e.g., Hindi, Tamil, Kannada) can broaden the application's reach beyond Telugu-speaking communities across different states.

7. Data Analytics for Public Health Insights

Aggregated, anonymized data collected through the application can be analyzed to identify community-level health trends, supporting district-level public health planning and targeted intervention design.

Role of AI in Healthcare Applications:

AI has the potential to significantly enhance digital healthcare by enabling intelligent symptom assessment, personalized reminders, predictive health monitoring, and automated analysis of health records. It is essential, however, that AI-powered features be implemented with appropriate safeguards and clearly communicated as assistive — not diagnostic — tools. Medical diagnosis and treatment decisions must always be made by licensed healthcare professionals.

CHAPTER 5: RECOMMENDATIONS

Based on the findings of the Community Service Project at Peddapanjani village and the development of the mobile healthcare application concept, the following recommendations are proposed to relevant stakeholders.

5.1 Recommendations for Community Residents

- Participate in periodic health camps organized by local health workers and visiting healthcare teams.
- Use medicine reminders — whether digital or written — to improve medication adherence, particularly for chronic conditions.
- Register for applicable government healthcare schemes (Ayushman Bharat, NHM) with assistance from the local ANM.
- Consult qualified healthcare professionals for diagnosis and treatment; do not rely solely on informal or unverified health information.

5.2 Recommendations for Healthcare Organizations

- Partner with technology developers to promote simple, accessible digital healthcare tools within rural communities.
- Conduct regular outreach camps in villages with limited health infrastructure, prioritizing maternal health, child immunization, and chronic disease management.
- Maintain updated information on healthcare facilities in publicly accessible formats to support digital locator features.

5.3 Recommendations for Software Developers

- Design rural healthcare applications with offline functionality as a core requirement, given limited internet connectivity in rural areas.
- Prioritize accessibility: large fonts, simple icons, and regional language support for users with varying digital literacy.
- Implement strong data privacy and security measures, given the sensitivity of personal health information.
- Conduct user testing with actual rural residents, including elderly users, before finalizing application design.

5.4 Recommendations for Government and Public Health Authorities

- Invest in last-mile internet connectivity to enable rural residents to benefit from digital healthcare services.
- Promote digital literacy programmes in schools and community centres to support adoption of mobile health tools.

- Encourage and fund collaborations between engineering colleges and rural healthcare authorities for community-oriented technology development.
- Regularize staffing at village-level health sub-centres to strengthen the primary healthcare foundation.

5.5 Recommendations for Educational Institutions

- Sustain and expand community service project programmes to deepen the connection between technical education and social responsibility.
- Encourage interdisciplinary collaborations between engineering, health sciences, and social science departments for community-oriented projects.
- Incorporate modules on healthcare technology, digital inclusion, and rural development into undergraduate CSE curricula.
- Celebrate and publish student CSP outcomes to motivate future cohorts and demonstrate the social value of engineering education.

Overall Recommendation

A well-designed mobile healthcare assistance application — developed with direct input from rural communities, built for offline use, and supported by local healthcare workers — has significant potential to improve healthcare awareness, promote preventive practices, and support better health outcomes in underserved rural communities. Sustained collaboration between technology developers, healthcare authorities, and educational institutions is essential to realizing this potential at scale.

CHAPTER 6: CONCLUSION

The Community Service Project at Peddapanjani village, Chittoor District, Andhra Pradesh provided the team with an invaluable opportunity to move beyond classroom learning and engage directly with a real community and its development challenges. Over a period of six weeks — from 10 May 2026 to 29 June 2026 — the team conducted a comprehensive socio-economic survey across 68 households, identified five key community problems related to healthcare access and awareness, organized a well-attended healthcare awareness programme for 65 residents, and developed a conceptual mini-project: the 'Mobile App for Rural Healthcare Assistance.'

The survey and field engagement confirmed that while basic infrastructure — electricity, mobile connectivity, and piped water — has reached the village, healthcare accessibility remains a significant challenge. The nearest government hospital is approximately 12 kilometres away, medication adherence is irregular, and awareness of both preventive healthcare practices and government healthcare entitlements is limited. The community awareness programme organized by the team was a meaningful step toward addressing these gaps, and community feedback directly shaped the features of the proposed mobile application.

The mini-project demonstrated how the team's technical training in Computer Science, Artificial Intelligence, and Machine Learning can be purposefully applied to address real-world social challenges. The proposed application — featuring digital health profiles, medicine reminders, a healthcare facility locator, emergency contacts, and health awareness content — represents a practical, community-validated digital solution with significant potential for real-world impact.

This Community Service Project reinforced the team's understanding that engineering is not merely a technical discipline but a social responsibility. The experience developed both professional competencies — including survey design, data analysis, technical documentation, and software architecture — and personal qualities including empathy, teamwork, cross-cultural communication, and civic awareness.

Learning Outcomes Achieved

- Applied software engineering and AI/ML principles to a real-world, community-validated healthcare problem.
- Developed proficiency in socio-economic survey design, data collection, and analysis.
- Strengthened professional communication skills through sustained direct community engagement.

- Organized and delivered a community awareness programme independently, from planning through execution.
- Gained first-hand experience of the socio-economic realities of rural India, deepening understanding of the community impact of technology.
- Strengthened teamwork, project management, and technical documentation capabilities.
- Recognized and internalized the social responsibility inherent in engineering education.

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STUDENT SELF-EVALUATION FOR THE COMMUNITY SERVICE PROJECT

Student Name: Narendra Kumar Paswan

Registration No.: 24781A33J7

Period of CSP: From: 10 May 2026 To: 29 June 2026

Date of Evaluation: 21 June 2026

#	Evaluation Criterion	Rating (1–5)
1	Oral communication	4
2	Written communication	4
3	Proactiveness	5
4	Interaction ability with community	4
5	Positive Attitude	5
6	Self-confidence	4
7	Ability to learn	5
8	Work Plan and organization	4
9	Professionalism	4
10	Creativity	4
11	Quality of work done	4
12	Time Management	5
13	Understanding the Community	4
14	Achievement of Desired Outcomes	4
15	OVERALL PERFORMANCE	4

Rating Scale: 1 = Very Poor, 2 = Poor, 3 = Average, 4 = Good, 5 = Excellent

Narendra Kumar Paswan — Signature of the Student

EVALUATION BY PERSON IN-CHARGE IN THE COMMUNITY**Student Name:** Narendra Kumar Paswan (Group Leader, Group No. 16)**Registration No.:** 24781A33J7**Period of CSP:** From: 10 May 2026 To: 29 June 2026**Date of Evaluation:** 21 June 2026**Name of the Person In-Charge:** Mr. K. Ravi Kumar**Designation:** Village Administrative Officer, Peddapanjani**Address:** Peddapanjani Village, Chittoor District, Andhra Pradesh

#	Evaluation Criterion	Rating (1–5)
1	Oral communication	4
2	Written communication	4
3	Proactiveness	5
4	Interaction ability with community	4
5	Positive Attitude	5
6	Self-confidence	5
7	Ability to learn	5
8	Work Plan and organization	4
9	Professionalism	4
10	Creativity	4
11	Quality of work done	4
12	Time Management	5
13	Understanding the Community	5
14	Achievement of Desired Outcomes	5
15	OVERALL PERFORMANCE	5

Rating Scale: 1 = Very Poor, 2 = Poor, 3 = Average, 4 = Good, 5 = Excellent

Mr. K. Ravi Kumar — Authorized Signatory with Seal

INTERNAL ASSESSMENT STATEMENT

Name of the Student: Narendra Kumar Paswan

Programme of Study: B.Tech – CSE (Artificial Intelligence & Machine Learning)

Year of Study: III Year B.Tech

Group No.: 16

Register No.: 24781A33J7

Name of the College: Sri Venkateswara College of Engineering and Technology
(Autonomous), Chittoor

University: Jawaharlal Nehru Technological University, Ananthapuramu

Sl.No	Evaluation Criterion	Maximum Marks	Marks Awarded
1	Activity Log	20	
2	Community Service Project Implementation	30	
3	Mini Project Work	25	
4	Oral Presentation	25	
	GRAND TOTAL	100	

_____ Mrs. T. Amruthavalli — Signature of Faculty Guide

Certified by:

_____ Signature of HoD / Principal with Seal